
Turn in 4

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MA 542 Section A1

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Note: there is a lot of explicit matrix computation involved in this assignment. I have included all computations at the end just incase you want to see them, but I wasn't sure and didn't want to make it too messy so like yeah.

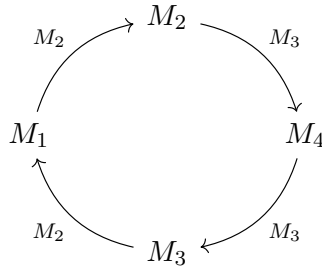
For simplicity, denote

$$M_1 := \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad M_2 := \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad M_3 := \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \quad M_4 := \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}.$$

Also let 1_R denote the identity matrix. Explicit computation yields that

$$M_1 M_2 = M_2, \quad M_3 M_2 = M_4, \quad M_4 M_3 = M_3, \quad M_2 M_3 = M_1.$$

This can be visualized with the diagram:



In some cases the operation is left-multiplication, and in others the operation is right-multiplication. The important thing to note is that if there exists some $i \in \{1, \dots, 4\}$ with $M_i \in I$, then it follows that $M_i \in I$ for all i , because I is an ideal and thus has the absorption property, and all M_i can be represented as a product in R on a single M_i by the above diagram. I will now show the ideal contains at least one of these.

Let

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in I$$

Since $A \neq 0$, there exists at least one of a, b, c, d which is nonzero. Explicit computation yields that

$$M_1 A M_1 = a M_1;$$

$$M_2 A M_2 = c M_2;$$

$$M_3 A M_3 = b M_3;$$

$$M_4 A M_4 = d M_4.$$

Since A is in the (two-sided) ideal, and all of these are products of A , we find that $a M_1, c M_2, b M_3, d M_4 \in I$. Since at least one of these scalar coefficients must be nonzero, we find that there exists at least

one nonzero $kM_i \in I$ for some nonzero $k \in \{a, b, c, d\}$. Since $kM_i \in I$, because I is an ideal, and $\frac{1}{k}1_R \in R$ because $k \neq 0$, we have

$$\frac{1}{k}1_R kM_i = 1_R M_i = M_i \in I.$$

Therefore, there exists at least one $M_i \in I$ by the absorption property. By the above logic, all M_i are thus in I .

Since I is an ideal, it must be a subring, and thus closed under addition. We thus have

$$M_1 + M_4 = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} = 1_R \in I.$$

Let $M \in \mathcal{M}_2(\mathbb{R})$ be any matrix. Then since $1_R \in I$, and I is an ideal, we have that $1_R M = M \in I$. Therefore, $\mathcal{M}_2(\mathbb{R}) \subseteq I$. Since I is an ideal, we have $I \subseteq \mathcal{M}_2(\mathbb{R})$, and thus $I = \mathcal{M}_2(\mathbb{R})$.

Computation!

$$M_1 M_2 = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad M_3 M_2 = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix},$$

$$M_4 M_3 = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \quad M_2 M_3 = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}.$$

$$M_1 A M_1 = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} a & b \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} a & 0 \\ 0 & 0 \end{pmatrix}.$$

$$M_2 A M_2 = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} c & d \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & c \\ 0 & 0 \end{pmatrix}.$$

$$M_3 A M_3 = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ a & b \end{pmatrix} \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ b & 0 \end{pmatrix}.$$

$$M_4 A M_4 = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ c & d \end{pmatrix} \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & d \end{pmatrix}.$$